

HW #2

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Quarto

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Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

Introduction

This document analyzes a restaurant inspections dataset. The analysis includes data loading, exploration, cleaning, and visualizations to answer several questions related to inspection scores, trends over time, comparisons by city, inspector, facility type, and a focused analysis on restaurants.

Setup

We start by loading the required libraries.

```
library(readr)    # for reading CSV files
library(dplyr)    # for data manipulation
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)  # for plotting
library(stringr)  # for string manipulation
```

Load the Data

Check the current working directory and load the dataset.

```
# Check current working directory
getwd()
```

```
[1] "/Users/SchoolAccount/Desktop/plan372_hm1repo/plan372hw2/hw2"
```

```
# Load the dataset
restaruant_data <- read_csv("/Users/SchoolAccount/Desktop/plan372_hm1repo/HWS/restaurant_insp")
```

```
Rows: 3875 Columns: 14
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr  (8): HSSID, DESCRIPTION, TYPE, INSPECTOR, NAME, RESTAURANTOPENDATE, CI...
```

```
dbl  (5): OBJECTID, SCORE, PERMITID, geocode_lon, geocode_lat
```

```
dtm (1): DATE_
```

```
i Use `spec()` to retrieve the full column specification for this data.  
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data Exploration

Examine the dataset by checking its dimensions, the first few rows, summary statistics, column names, and overall structure.

```
# Dimensions of the dataset: number of rows and columns  
dim(restaruant_data)
```

```
[1] 3875    14
```

```
# Display the first six rows of the dataset  
head(restaruant_data)
```

```
# A tibble: 6 x 14  
  OBJECTID HSISID SCORE DATE_ DESCRIPTION TYPE INSPECTOR PERMITID  
    <dbl> <chr> <dbl> <dtm>      <chr>      <chr> <chr>      <dbl>  
1 25137654 04092~ 97 2017-10-22 04:00:00 <NA>      Insp~ Karla Cr~ 13405  
2 25115128 04092~ 96 2019-02-27 05:00:00 "*Notice* ~ Insp~ Meghan S~ 13939  
3 25123164 04092~ 98.5 2019-03-04 05:00:00 "*NOTICE* ~ Insp~ Kaitlyn ~ 20554  
4 25128895 04092~ 90.5 2019-03-23 04:00:00 "Opening c~ Insp~ Angela M~ 15506  
5 25124786 04092~ 97.5 2019-04-24 04:00:00 "*NOTICE* ~ Insp~ Patricia~ 14839  
6 25108274 04092~ 98 2019-05-14 04:00:00 "*NOTICE* ~ Insp~ Maria Po~ 8851  
# i 6 more variables: NAME <chr>, RESTAURANTOPENDATE <chr>, CITY <chr>,  
# FACILITYTYPE <chr>, geocode_lon <dbl>, geocode_lat <dbl>
```

```
# Summary of numeric columns  
summary(restaruant_data)
```

OBJECTID	HSISID	SCORE
Min. :25091064	Length:3875	Min. : 0.00
1st Qu.:25102773	Class :character	1st Qu.: 95.50
Median :25118689	Mode :character	Median : 97.50
Mean :25116443		Mean : 96.92
3rd Qu.:25129320		3rd Qu.: 98.50

Max. :25139899 Max. :100.00

DATE_	DESCRIPTION	TYPE
Min. :2017-10-22 04:00:00.00	Length:3875	Length:3875
1st Qu.:2021-08-20 04:00:00.00	Class :character	Class :character
Median :2021-10-28 04:00:00.00	Mode :character	Mode :character
Mean :2021-09-26 21:08:39.32		
3rd Qu.:2021-12-14 05:00:00.00		
Max. :2022-01-31 05:00:00.00		

INSPECTOR	PERMITID	NAME	RESTAURANTOPENDATE
Length:3875	Min. : 1	Length:3875	Length:3875
Class :character	1st Qu.: 6136	Class :character	Class :character
Mode :character	Median :12872	Mode :character	Mode :character
	Mean :12285		
	3rd Qu.:18374		
	Max. :23602		

CITY	FACILITYTYPE	geocode_lon	geocode_lat
Length:3875	Length:3875	Min. : -78.94	Min. : 0.00
Class :character	Class :character	1st Qu.: -78.78	1st Qu.: 35.75
Mode :character	Mode :character	Median : -78.66	Median : 35.80
		Mean : -77.74	Mean : 35.37
		3rd Qu.: -78.60	3rd Qu.: 35.85
		Max. : 0.00	Max. : 36.05
		NA's :296	NA's :296

```
# Names of the columns
names(restaruant_data)
```

```
[1] "OBJECTID"      "HSISID"        "SCORE"
[4] "DATE_"         "DESCRIPTION"    "TYPE"
[7] "INSPECTOR"     "PERMITID"       "NAME"
[10] "RESTAURANTOPENDATE" "CITY"          "FACILITYTYPE"
[13] "geocode_lon"   "geocode_lat"
```

```
# Structure of the dataset (includes types of each column)
str(restaruant_data)
```

```
spc_tbl_ [3,875 x 14] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ OBJECTID      : num [1:3875] 25137654 25115128 25123164 25128895 25124786 ...
```

```

$ HSISID          : chr [1:3875] "04092021735" "04092013943" "04092021636" "04092021737"
$ SCORE           : num [1:3875] 97 96 98.5 90.5 97.5 98 100 99.5 100 90.5 ...
$ DATE_           : POSIXct[1:3875], format: "2017-10-22 04:00:00" "2019-02-27 05:00:00"
$ DESCRIPTION     : chr [1:3875] NA "*Notice* Effective January 1, 2019, the NC Food Code
$ TYPE           : chr [1:3875] "Inspection" "Inspection" "Inspection" "Inspection" ...
$ INSPECTOR       : chr [1:3875] "Karla Crowder" "Meghan Scott" "Kaitlyn Yow" "Angela Myer
$ PERMITID        : num [1:3875] 13405 13939 20554 15506 14839 ...
$ NAME            : chr [1:3875] "THE CENTERLINE CAFE" "St. Raphael Hall Foodservice" "KN
$ RESTAURANTOPENDATE: chr [1:3875] "2013/02/01 05:00:00+00" "2003/12/01 05:00:00+00" "2010/
$ CITY            : chr [1:3875] "CARY" "RALEIGH" "KNIGHTDALE" "CARY" ...
$ FACILITYTYPE    : chr [1:3875] "Food Stand" "Restaurant" "Food Stand" "Food Stand" ...
$ geocode_lon     : num [1:3875] -78.8 -78.6 -78.5 -78.8 0 ...
$ geocode_lat     : num [1:3875] 35.8 35.9 35.8 35.8 0 ...
- attr(*, "spec")=
.. cols(
..   OBJECTID = col_double(),
..   HSISID = col_character(),
..   SCORE = col_double(),
..   DATE_ = col_datetime(format = ""),
..   DESCRIPTION = col_character(),
..   TYPE = col_character(),
..   INSPECTOR = col_character(),
..   PERMITID = col_double(),
..   NAME = col_character(),
..   RESTAURANTOPENDATE = col_character(),
..   CITY = col_character(),
..   FACILITYTYPE = col_character(),
..   geocode_lon = col_double(),
..   geocode_lat = col_double()
.. )
- attr(*, "problems")=<externalptr>

```

Data Cleaning

Count missing values per column, filter out rows with missing values in vital columns (CITY, FACILITYTYPE, SCORE), and check for duplicate rows.

```

# Count missing data in each column
colSums(is.na(restaruant_data))

```

OBJECTID

HSISID

SCORE

DATE_

0	0	0	0
DESCRIPTION	TYPE	INSPECTOR	PERMITID
516	0	0	0
NAME RESTAURANTOPENDATE		CITY	FACILITYTYPE
296	296	296	296
geocode_lon	geocode_lat		
296	296		

```
# Filter out rows with missing values in CITY, FACILITYTYPE, and SCORE
restaruant_data <- restaruant_data %>%
  filter(!is.na(CITY) & !is.na(FACILITYTYPE) & !is.na(SCORE))

# Check the dimensions after filtering
dim(restaruant_data)
```

```
[1] 3579  14
```

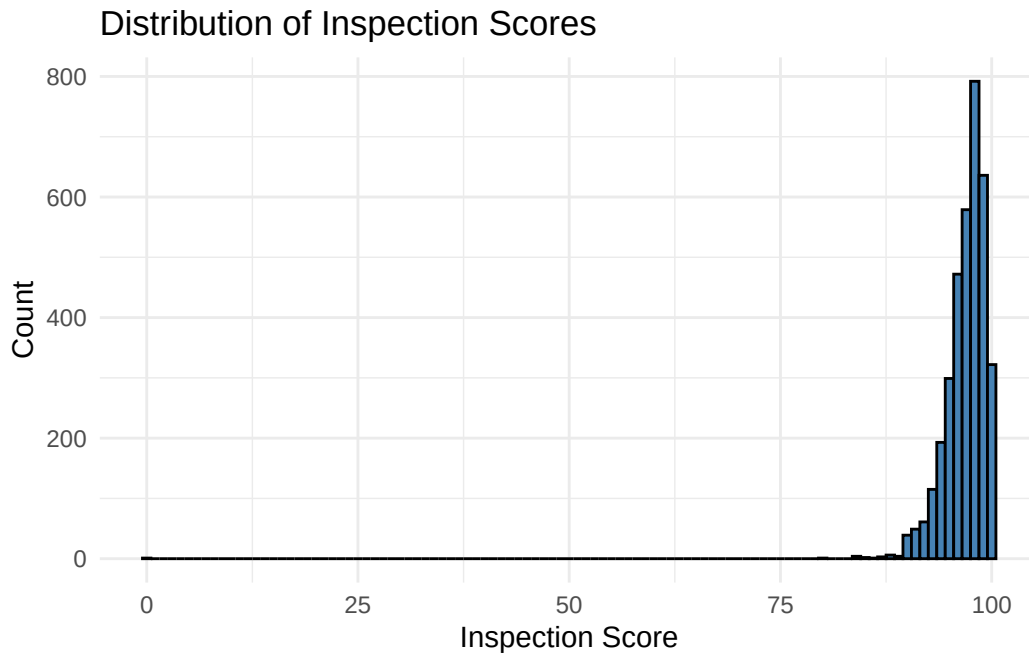
```
# Check for any duplicate rows (should return 0 if there are none)
sum(duplicated(restaruant_data))
```

```
[1] 0
```

Question 1: Distribution of Inspection Scores

Visualize the overall distribution of inspection scores using a histogram.

```
ggplot(restaruant_data, aes(x = SCORE)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Inspection Scores",
       x = "Inspection Score",
       y = "Count") +
  theme_minimal()
```



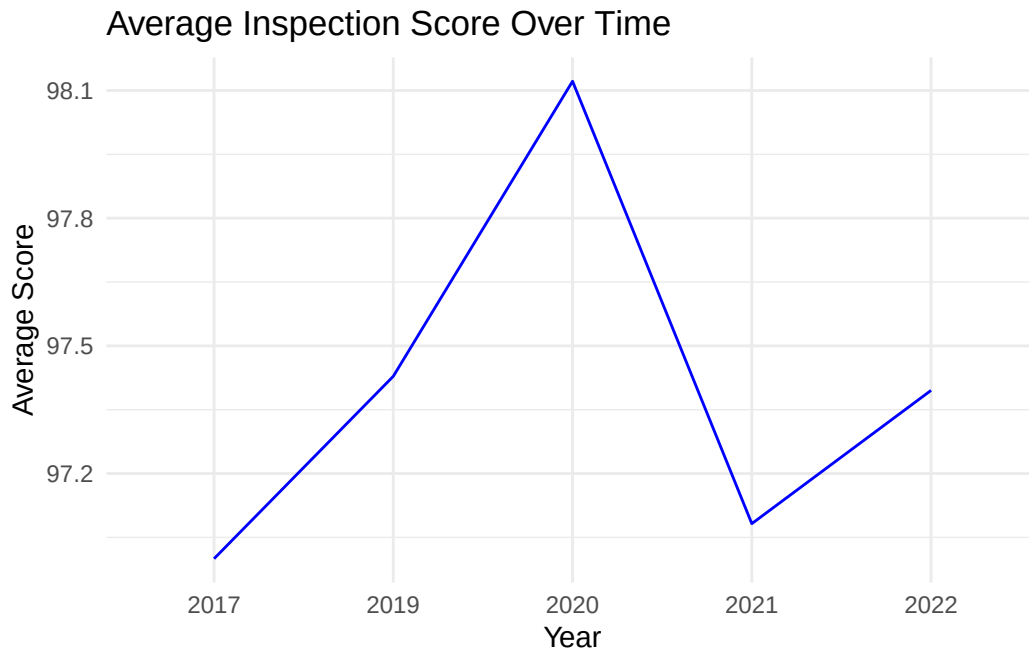
Question 2: Trends in Inspection Scores Over Time

Investigate whether there are trends over time by extracting the year from the `DATE_` column and calculating the average inspection score per year.

```
# Extract the year from the DATE_ column
restaruant_data$INSPECTION_YEAR <- format(restaruant_data$DATE_, "%Y")

# Group by year and calculate the average inspection score
avg_score_by_year <- restaruant_data %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE))

# Plot the trend over the years
ggplot(avg_score_by_year, aes(x = INSPECTION_YEAR, y = Average_Score, group = 1)) +
  geom_line(color = "blue") +
  labs(title = "Average Inspection Score Over Time",
       x = "Year",
       y = "Average Score") +
  theme_minimal()
```



Scores peaked in 2020 and were at their lowest in 2017.

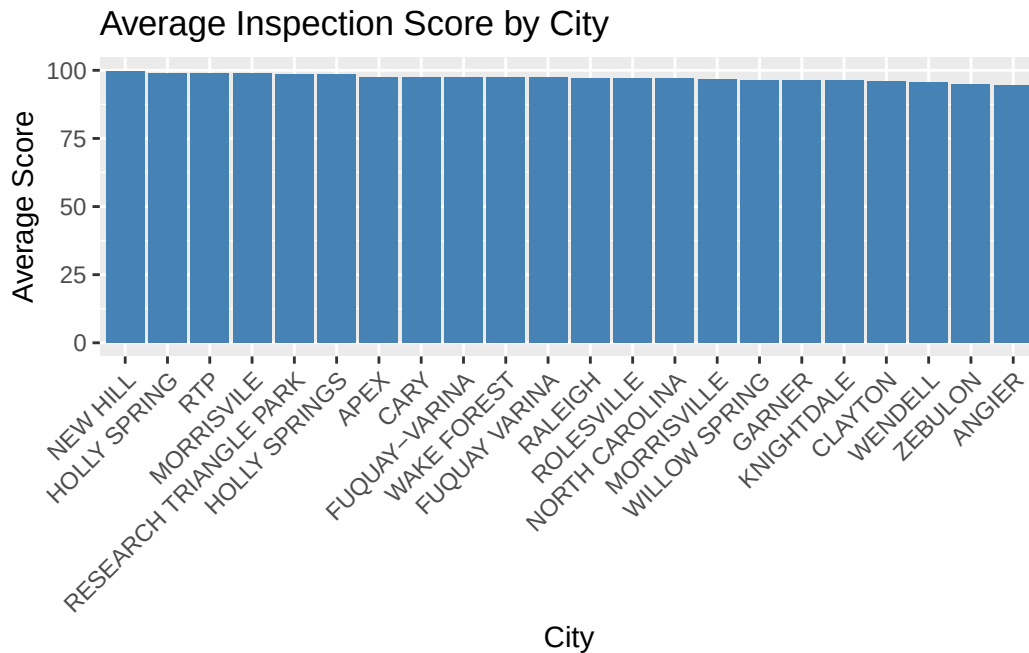
Question 3: Inspection Scores by City

Check if inspection scores vary by city by standardizing city names and calculating average scores.

```
# Standardize CITY names to uppercase
restaruant_data$CITY <- str_to_upper(restaruant_data$CITY)

# Group by CITY and calculate average inspection score and count
avg_score_by_city <- restaruant_data %>%
  group_by(CITY) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE), Count = n())

# Create a bar plot for average scores by city
ggplot(avg_score_by_city, aes(x = reorder(CITY, -Average_Score), y = Average_Score)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Average Inspection Score by City",
       x = "City",
       y = "Average Score") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Inspection scores do not vary by city.

Question 4: Inspection Scores by Inspector

Summarize the average score and the number of inspections for each inspector.

```
inspector_summary <- restaruant_data %>%
  group_by(INSPECTOR) %>%
  summarize(
    Average_Score = mean(SCORE, na.rm = TRUE),
    Inspection_Count = n()
  ) %>%
  arrange(desc(Average_Score))
```

```
# Display the inspector summary table
inspector_summary
```

```
# A tibble: 39 x 3
```

	INSPECTOR	Average_Score	Inspection_Count
	<chr>	<dbl>	<int>
1	James Smith	99	1
2	Greta Welch	98.5	2

3	Kaitlyn Yow	98.5	1
4	Jason Dunn	98.5	61
5	John Wulffert	98.4	154
6	Jamie Phelps	98.2	149
7	Nicole Millard	98.1	146
8	Zachary Carter	98.1	150
9	Brittany Thomas	98	3
10	Dipatrimarki Farkas	97.8	155

i 29 more rows

Inspection scores basically do not vary by inspector.

Question 5: Sample Size Considerations

Examine the sample sizes for each group (city, inspector, and year) to assess if small sample sizes might affect the results.

```
# Sample size by city
avg_score_by_city %>%
  select(CITY, Count) %>%
  arrange(Count)
```

```
# A tibble: 22 x 2
  CITY          Count
  <chr>         <int>
1 ANGIER          1
2 HOLLY SPRING    1
3 NORTH CAROLINA  1
4 RESEARCH TRIANGLE PARK 1
5 RTP             1
6 MORRISVILLE   2
7 NEW HILL        2
8 WILLOW SPRING   2
9 CLAYTON         4
10 ROLESVILLE   24
# i 12 more rows
```

```
# Inspectors with less than 5 inspections
inspector_summary %>%
  filter(Inspection_Count < 5)
```

```
# A tibble: 5 x 3
  INSPECTOR      Average_Score Inspection_Count
  <chr>          <dbl>          <int>
1 James Smith      99              1
2 Greta Welch     98.5             2
3 Kaitlyn Yow      98.5             1
4 Brittney Thomas  98              3
5 Thomas Jumalon   89              3
```

```
# Sample size by year
yearly_sample_size <- restaruant_data %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Inspection_Count = n()) %>%
  arrange(INSPECTION_YEAR)

# Display the sample size by year
yearly_sample_size
```

```
# A tibble: 5 x 2
  INSPECTION_YEAR Inspection_Count
  <chr>          <int>
1 2017              1
2 2019             28
3 2020             37
4 2021           2853
5 2022            660
```

There were a lot more inspections in certain cities than others (max 196, min 1). There were also a lot more inspections in the year 2021 (2853) than 2017 (1).

Question 6: Inspection Scores by Facility Type

Determine whether the scores for restaurants are higher than for other types of food-service facilities by grouping by FACILITYTYPE.

```
# Group by FACILITYTYPE and calculate average score and count
facility_summary <- restaruant_data %>%
  group_by(FACILITYTYPE) %>%
  summarize(
    Average_Score = mean(SCORE, na.rm = TRUE),
```

```

    Inspection_Count = n()
  ) %>%
  arrange(desc(Average_Score))

# View the facility summary table
facility_summary

```

```

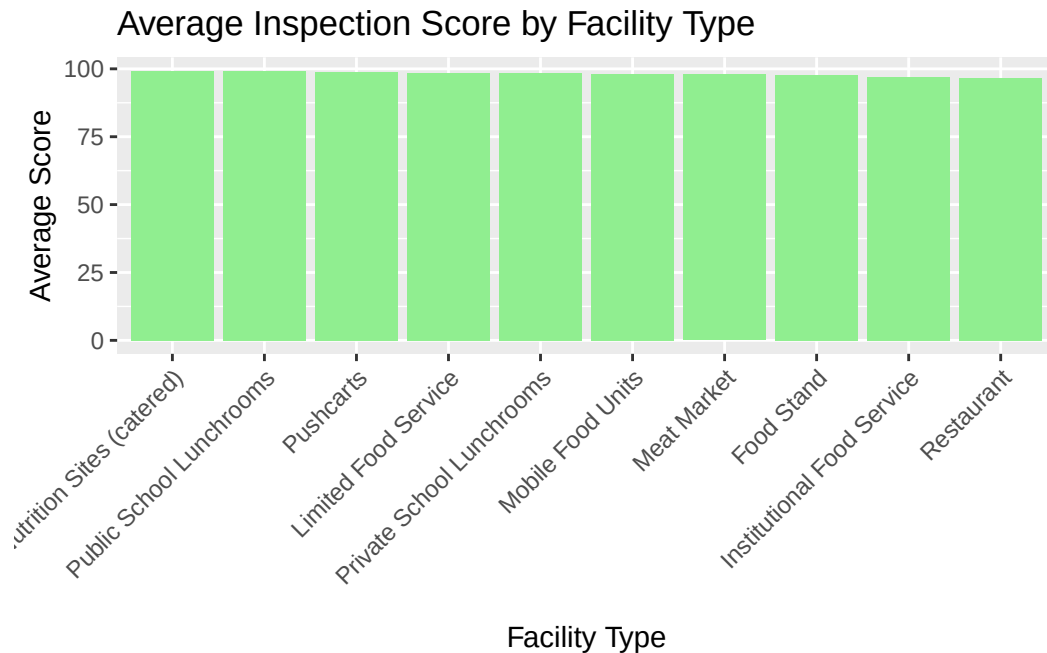
# A tibble: 10 x 3
  FACILITYTYPE      Average_Score Inspection_Count
  <chr>           <dbl>         <int>
1 Elderly Nutrition Sites (catered)    99.2             8
2 Public School Lunchrooms            99.2            185
3 Pushcarts                          98.8             39
4 Limited Food Service                 98.5              1
5 Private School Lunchrooms           98.5             13
6 Mobile Food Units                   98.1            181
7 Meat Market                        98.0             93
8 Food Stand                         97.7            661
9 Institutional Food Service          96.9             46
10 Restaurant                        96.7           2352

```

```

# Bar plot of average score by facility type
ggplot(facility_summary, aes(x = reorder(FACILITYTYPE, -Average_Score), y = Average_Score)) +
  geom_bar(stat = "identity", fill = "lightgreen") +
  labs(title = "Average Inspection Score by Facility Type",
       x = "Facility Type",
       y = "Average Score") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



All of the facilities are about the same. Nutrition sites and school lunchrooms generally score much better than restaurants.

Question 7: Analyses for Restaurants Specifically

Repeat the analyses above for restaurants only by first filtering the dataset.

```
# Filter the data for restaurants only
restaruant_only <- restaruant_data %>%
  filter(FACILITYTYPE == "Restaurant")

# Check the number of observations for restaurants
nrow(restaruant_only)
```

```
[1] 2352
```

7.1: Histogram of Inspection Scores for Restaurants

```
ggplot(restaruant_only, aes(x = SCORE)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Inspection Scores for Restaurants",
       x = "Inspection Score",
       y = "Count") +
  theme_minimal()
```



7.2: Inspection Trends by Year for Restaurants

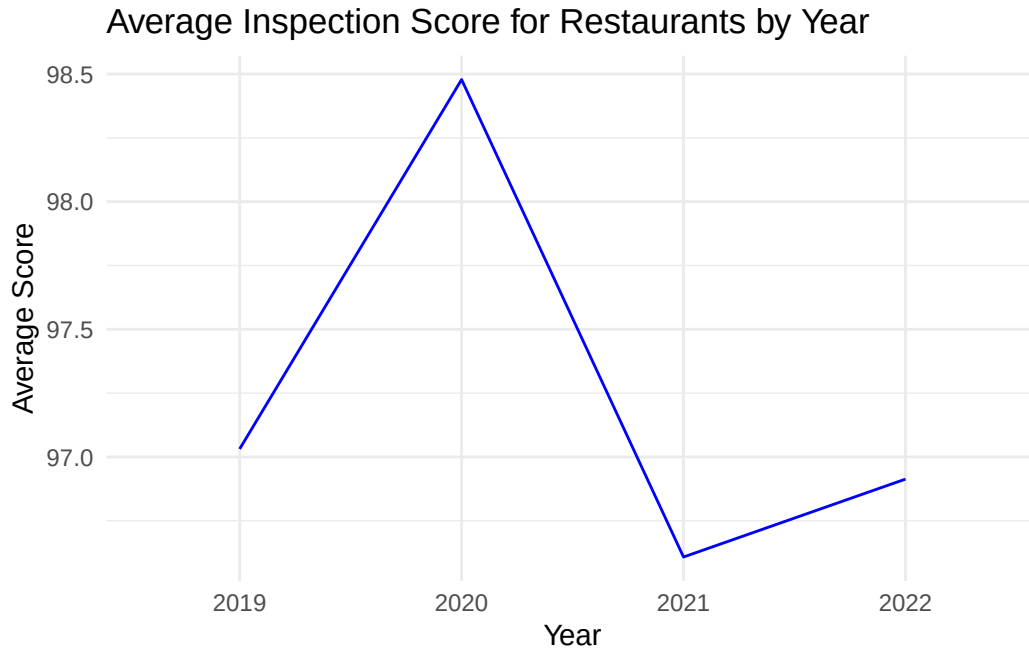
Extract the year for restaurants and plot the average inspection score by year.

```
# Extract the year from the DATE_ column for restaurants
restaruant_only$INSPECTION_YEAR <- format(restaruant_only$DATE_, "%Y")

# Group by year and calculate the average score for restaurants
avg_score_restaurants_by_year <- restaruant_only %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE))

# Plot the trend over the years for restaurants
ggplot(avg_score_restaurants_by_year, aes(x = INSPECTION_YEAR, y = Average_Score, group = 1)) +
  geom_line(color = "blue") +
```

```
labs(title = "Average Inspection Score for Restaurants by Year",
     x = "Year",
     y = "Average Score") +
theme_minimal()
```



Peaked in 2020 and lowest in 2021.

7.3: Inspection Scores by City for Restaurants

```
# Group by CITY and calculate average score and count for restaurants
avg_score_restaurants_by_city <- restaurant_only %>%
  group_by(CITY) %>%
  summarize(Average_Score = mean(SCORE, na.rm = TRUE), Count = n())

# Display the table for restaurants by city
avg_score_restaurants_by_city
```

```
# A tibble: 21 x 3
  CITY          Average_Score Count
<chr>          <dbl> <int>
1 ANGIER          94.5     1
```

```

2 APEX                97.1    108
3 CARY                97.3    406
4 CLAYTON             93       1
5 FUQUAY VARINA       96.9     49
6 FUQUAY-VARINA       97.0     27
7 GARNER              95.8     93
8 HOLLY SPRING        99       1
9 HOLLY SPRINGS       98.0     79
10 KNIGHTDALE         95.1     49
# i 11 more rows

```

Lot more inspections in the cities with higher scores.

7.4: Inspection Scores by Inspector for Restaurants

```

# Group by INSPECTOR and calculate average score and inspection count for restaurants
inspector_summary_restaurants <- restaruant_only %>%
  group_by(INSPECTOR) %>%
  summarize(
    Average_Score = mean(SCORE, na.rm = TRUE),
    Inspection_Count = n()
  ) %>%
  arrange(desc(Average_Score))

# Display the inspector summary table for restaurants
inspector_summary_restaurants

```

```

# A tibble: 38 x 3
  INSPECTOR      Average_Score Inspection_Count
  <chr>          <dbl>          <int>
1 James Smith    99              1
2 Greta Welch    98.5            2
3 John Wulffert  98.1           111
4 Brittny Thomas 98              3
5 Jamie Phelps   97.9           108
6 Zachary Carter 97.8           100
7 Dipatrimarki Farkas 97.7           118
8 Loc Nguyen     97.6            74
9 Ginger Johnson 97.6            35
10 Nicole Millard 97.6            48
# i 28 more rows

```


No major outliers, all about the same.

7.5: Sample Size Check for Restaurants

Examine sample sizes by city, inspector, and year for the restaurant-only subset.

```
# Sample size by city for restaurants
avg_score_restaurants_by_city %>%
  select(CITY, Count) %>%
  arrange(Count)
```

```
# A tibble: 21 x 2
  CITY                Count
  <chr>              <int>
1 ANGIER                1
2 CLAYTON                1
3 HOLLY SPRING          1
4 MORRISVILLE          1
5 NEW HILL               1
6 RESEARCH TRIANGLE PARK 1
7 RTP                   1
8 WILLOW SPRING          1
9 ROLESVILLE           13
10 WENDELL               20
# i 11 more rows
```

```
# Inspectors with fewer than 5 inspections among restaurants
inspector_summary_restaurants %>%
  filter(Inspection_Count < 5)
```

```
# A tibble: 5 x 3
  INSPECTOR      Average_Score Inspection_Count
  <chr>          <dbl>          <int>
1 James Smith      99              1
2 Greta Welch     98.5             2
3 Brittney Thomas  98              3
4 Jason Dunn      97.1             4
5 Thomas Jumalon   88              2
```

```
# Sample size by year for restaurants
yearly_sample_size_restaurants <- restaruant_only %>%
  group_by(INSPECTION_YEAR) %>%
  summarize(Inspection_Count = n()) %>%
  arrange(INSPECTION_YEAR)
```

```
# Display the sample size by year for restaurants
yearly_sample_size_restaurants
```

```
# A tibble: 4 x 2
  INSPECTION_YEAR Inspection_Count
  <chr>           <int>
1 2019             16
2 2020             23
3 2021          1916
4 2022            397
```

Big differences in number of inspections in different cities and the different years.

Github repo: <https://github.com/davisdickinson/plan372hw2.git>